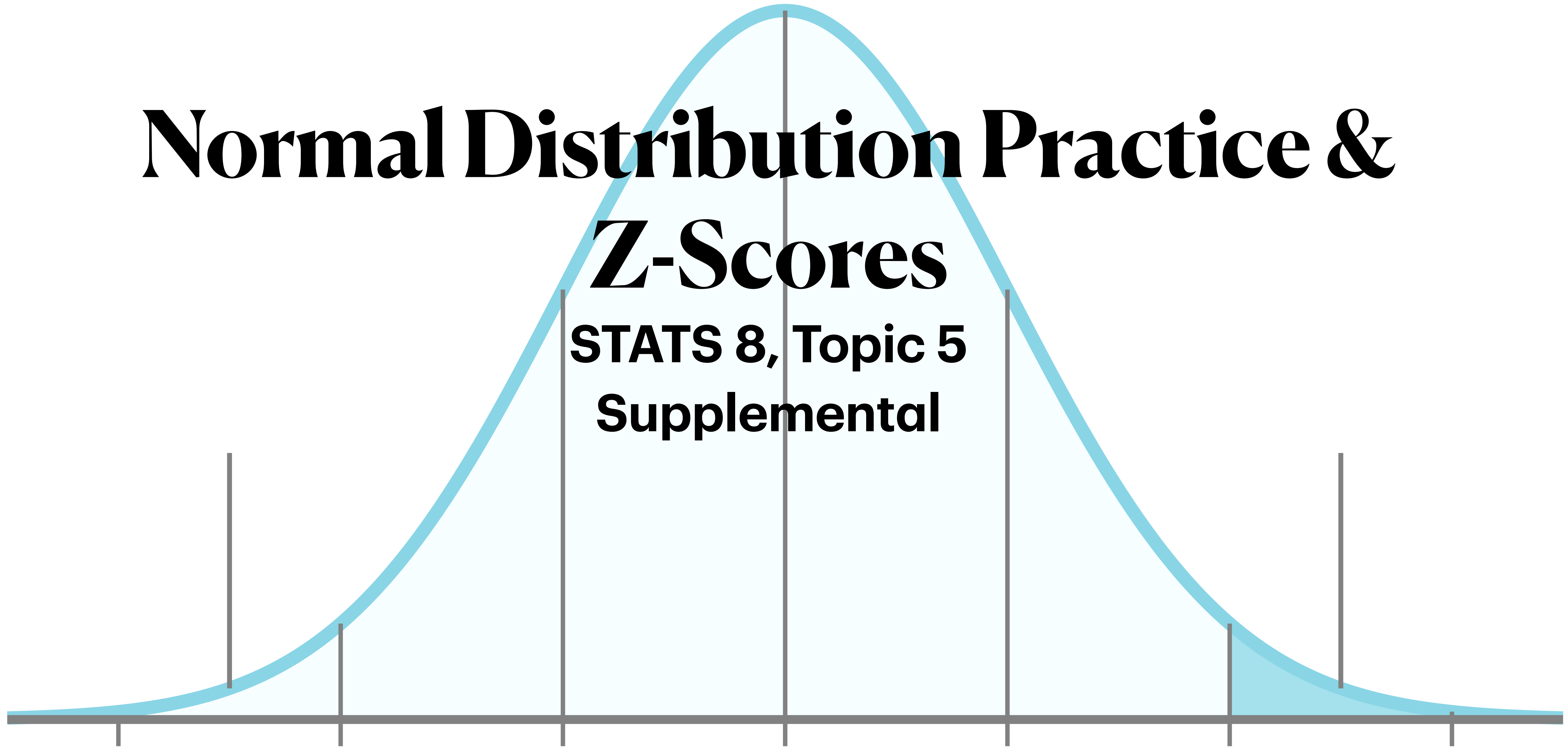


Normal Distribution Practice & Z-Scores

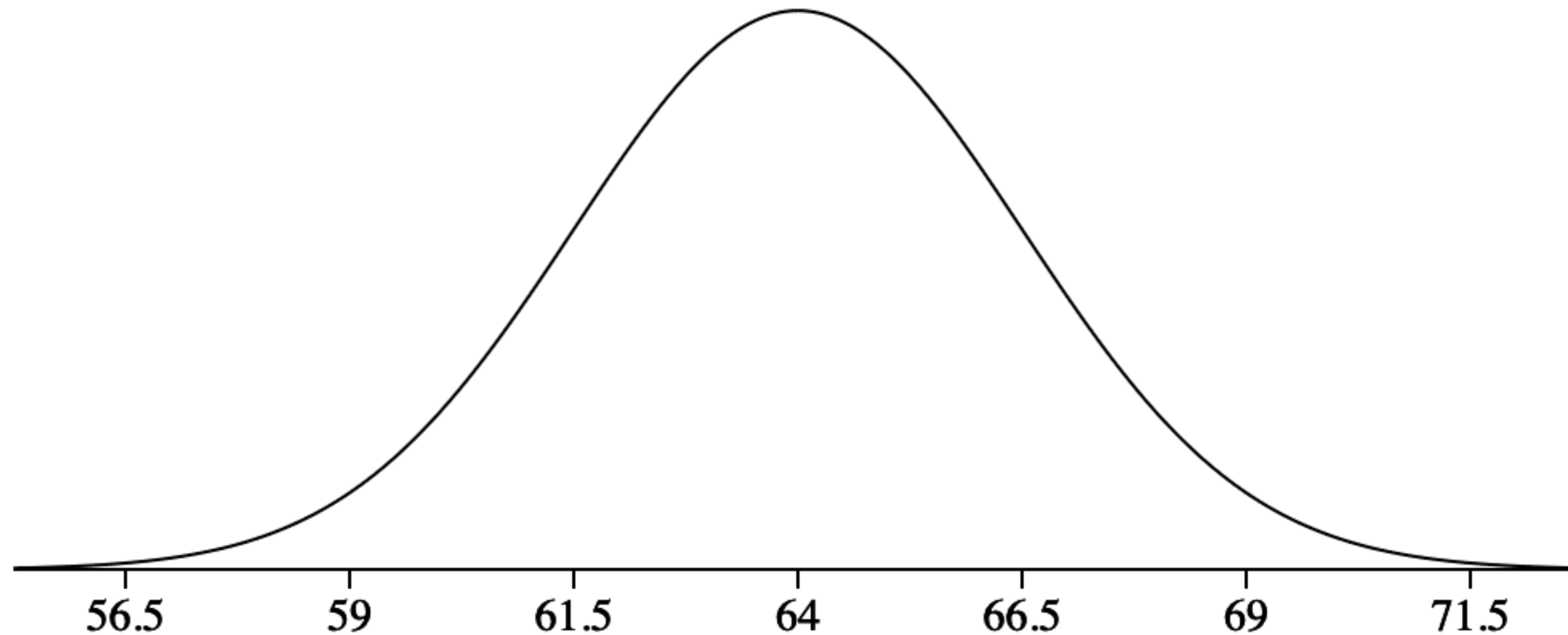
**STATS 8, Topic 5
Supplemental**



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

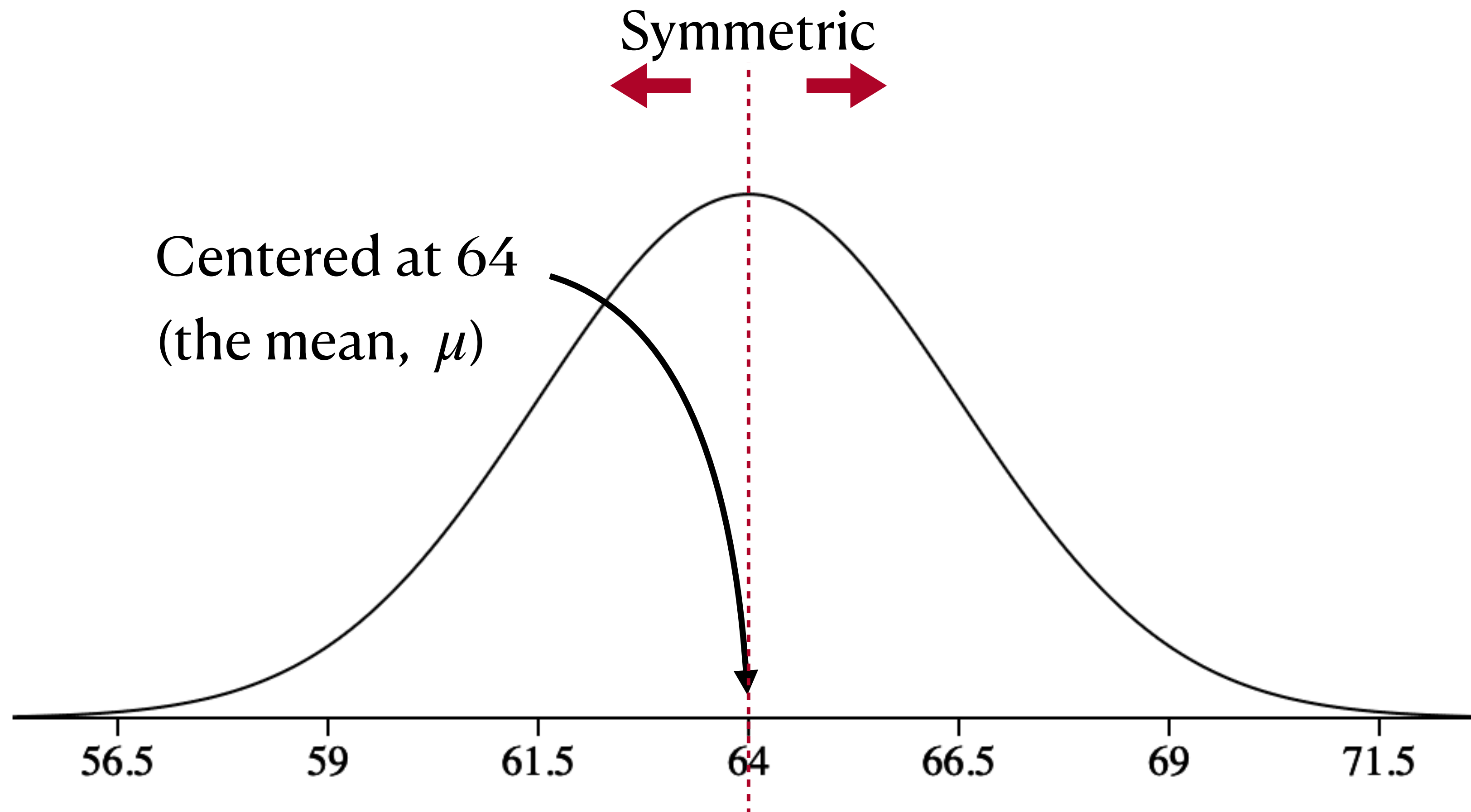
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution.



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

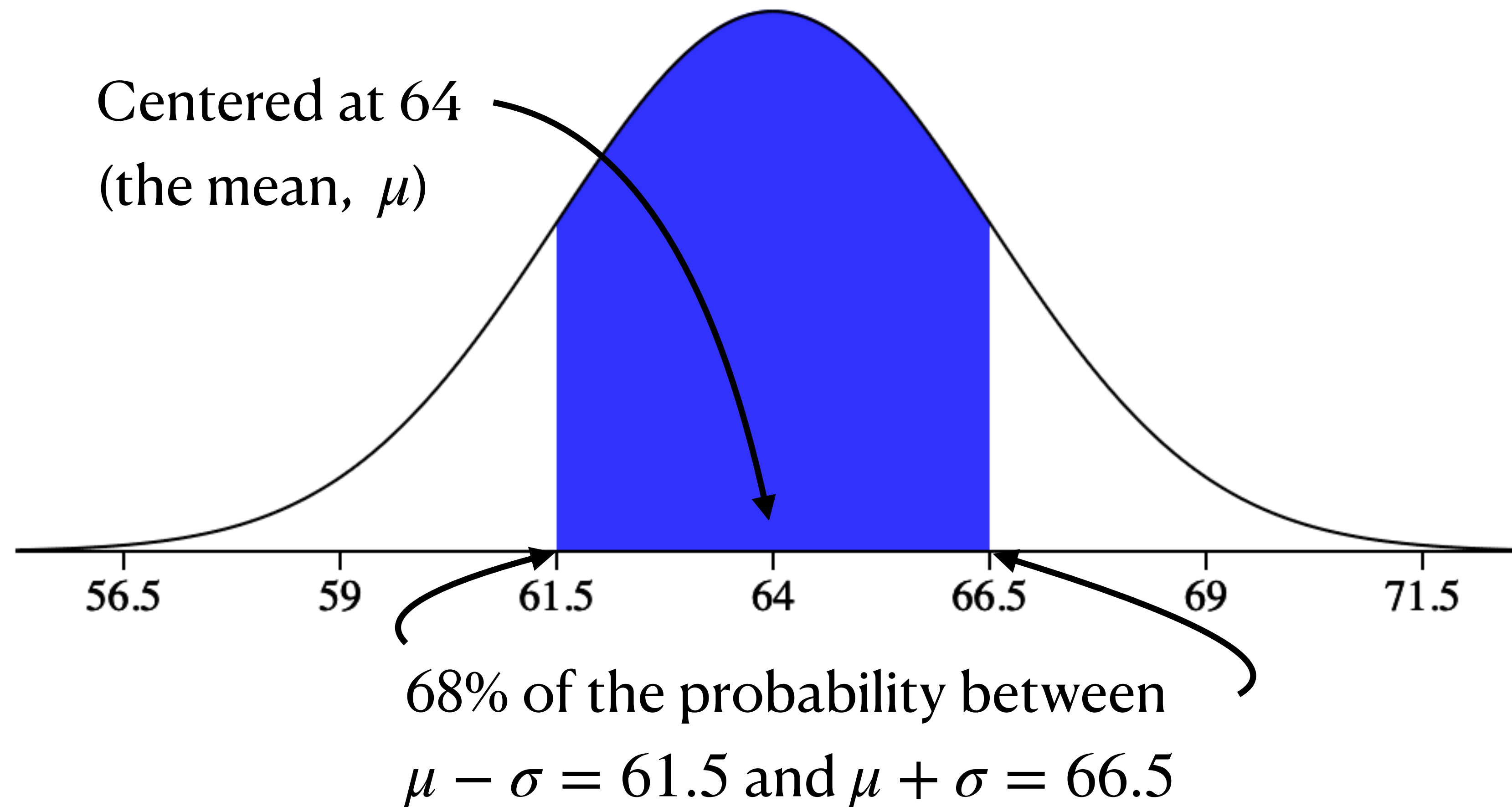
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution.



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

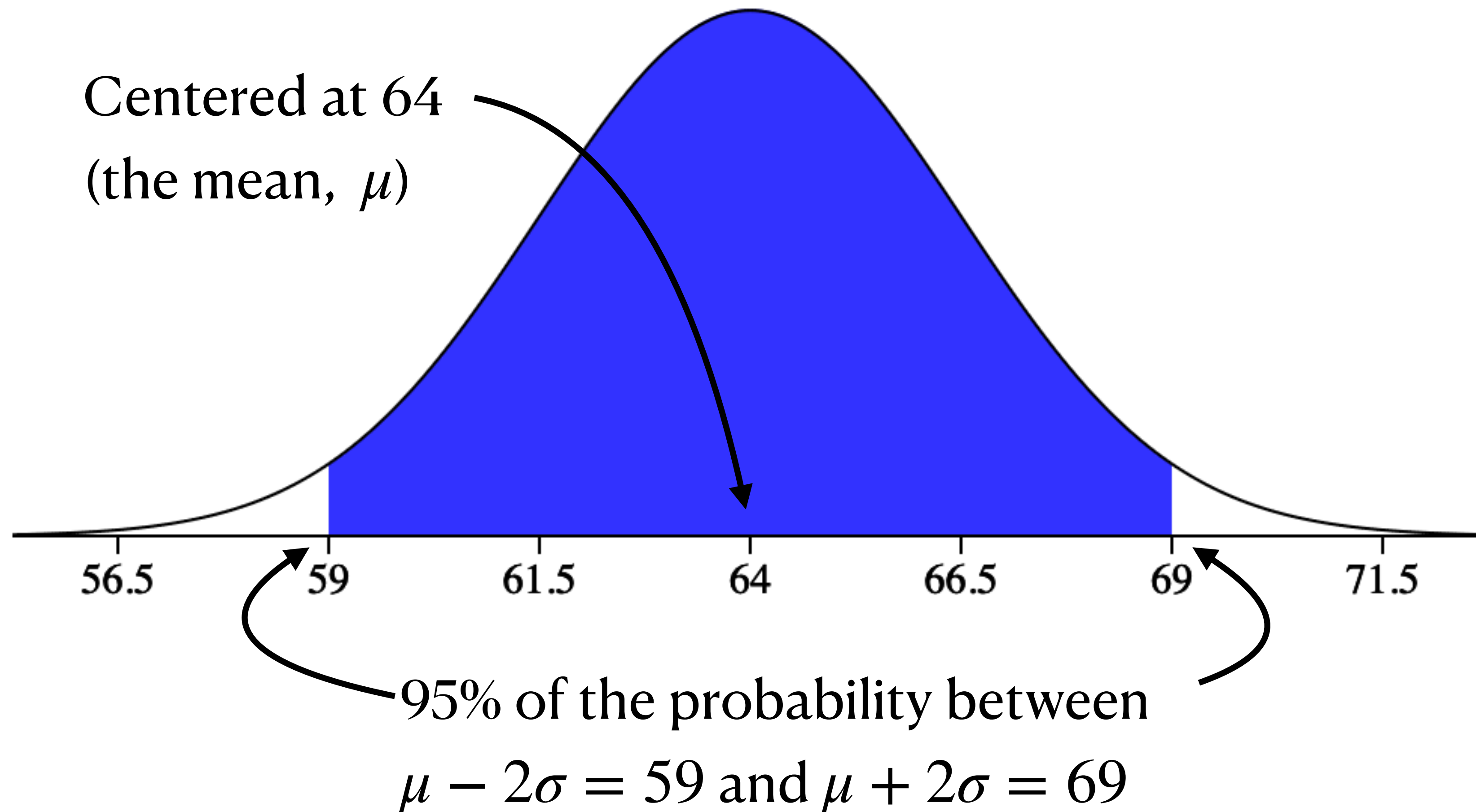
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution.



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

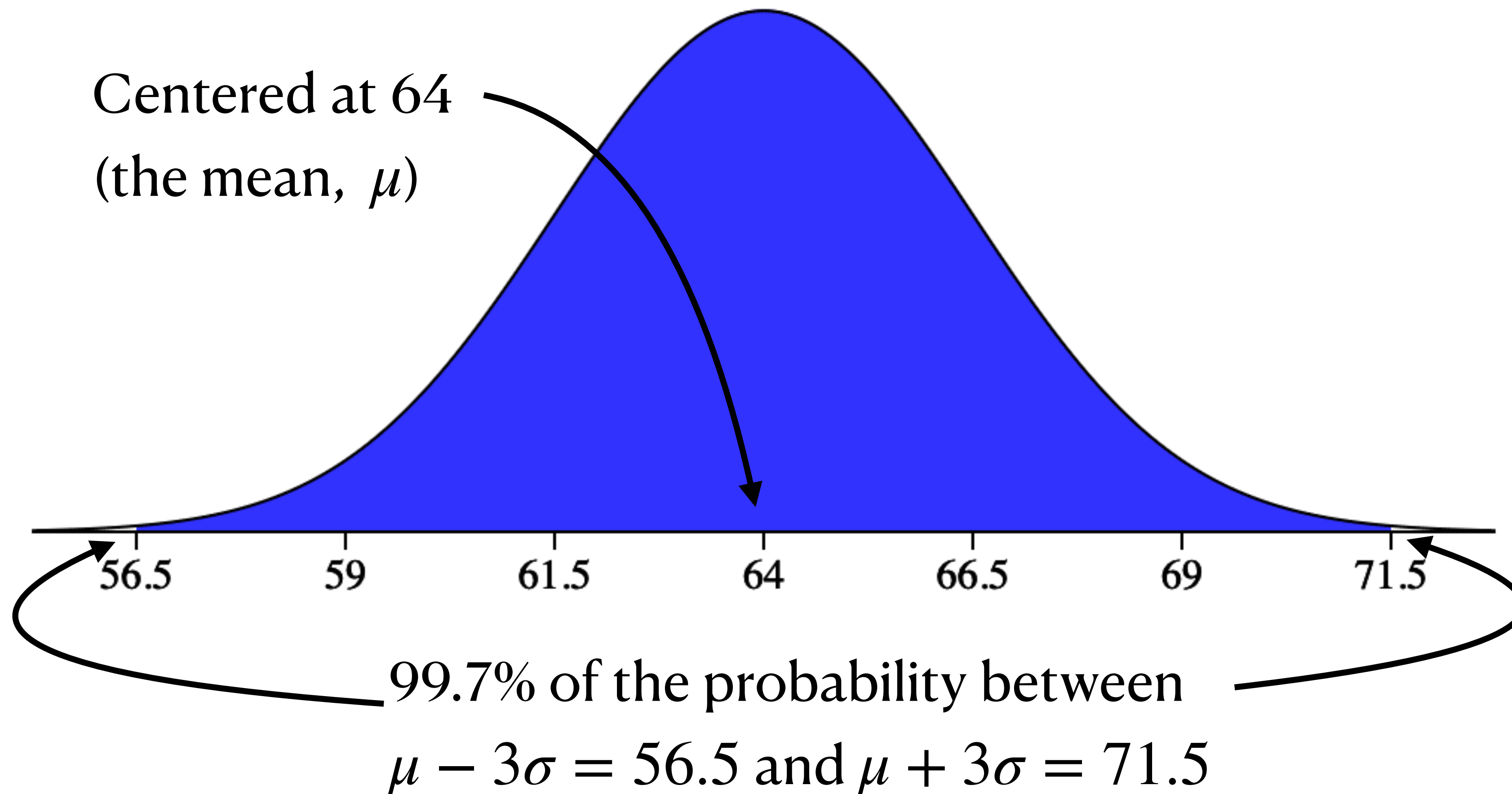
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution.



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

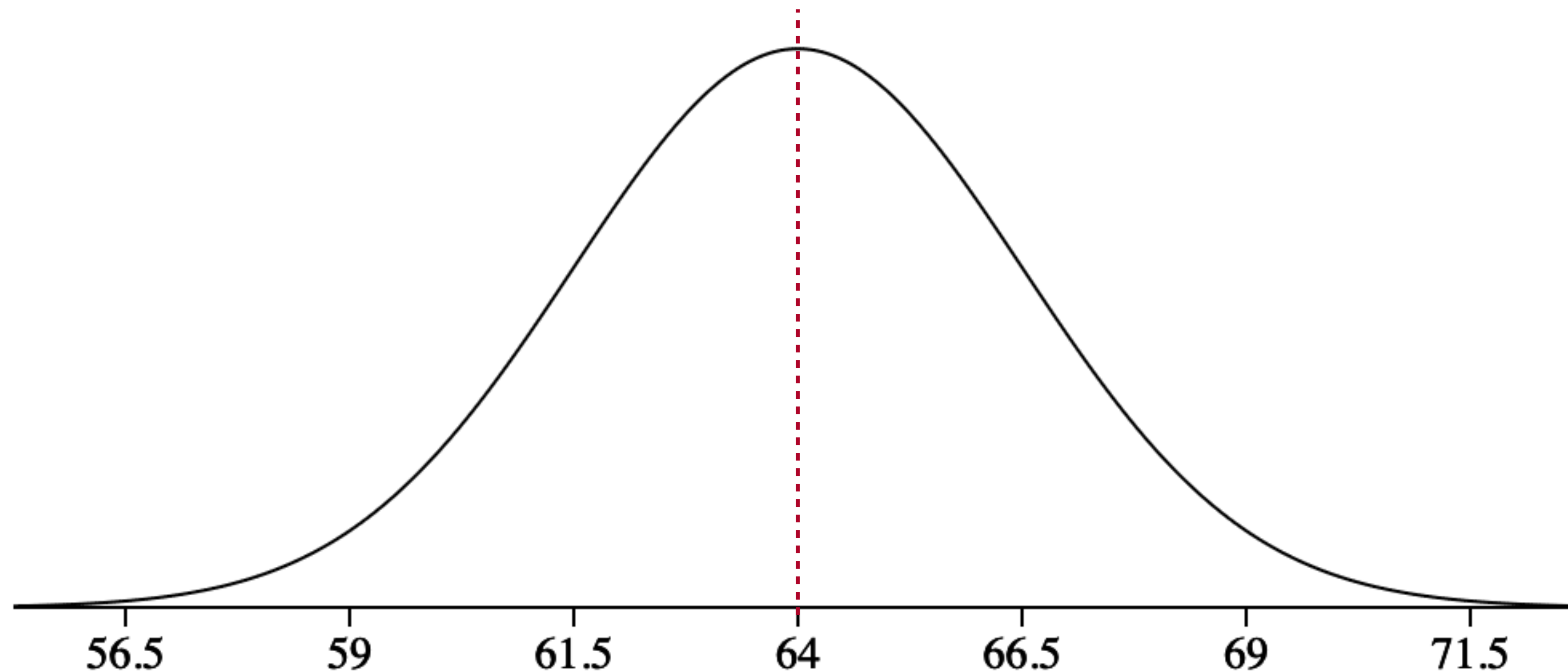
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution.



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

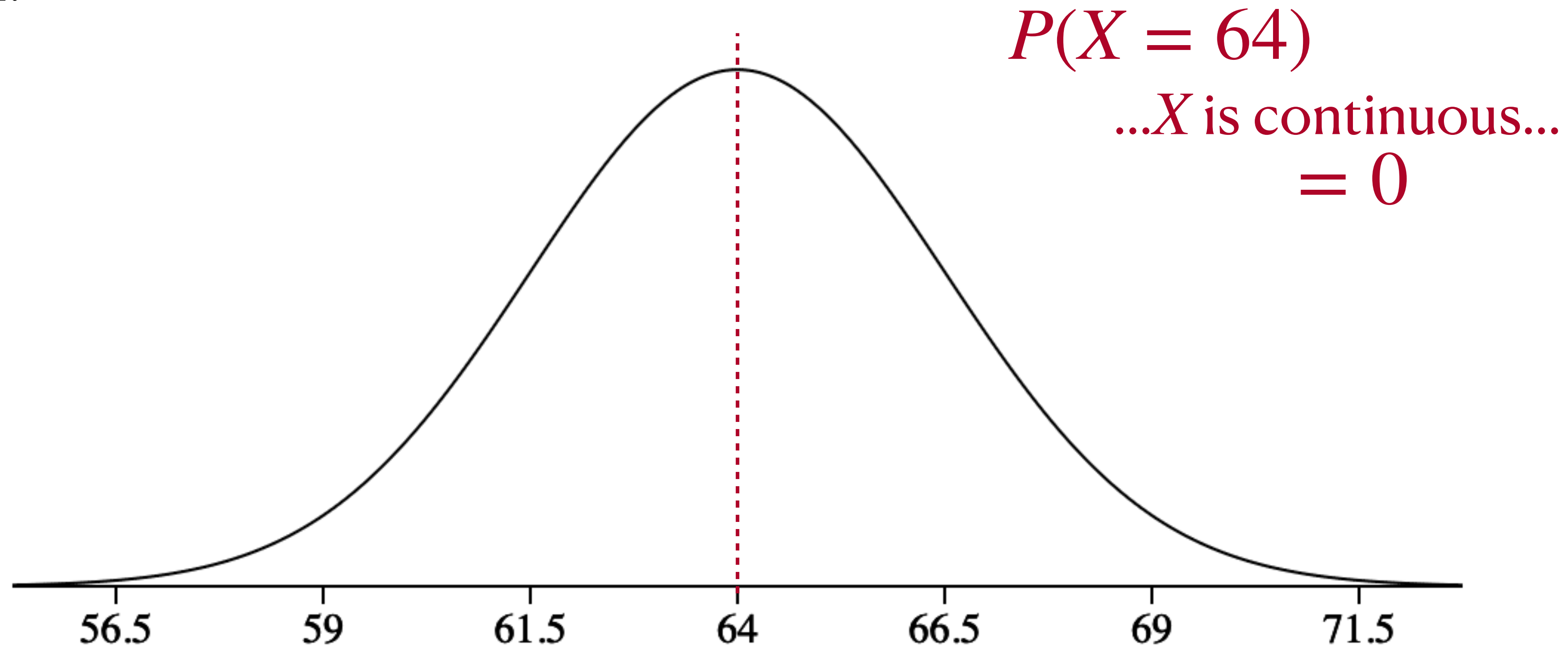
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is 64 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

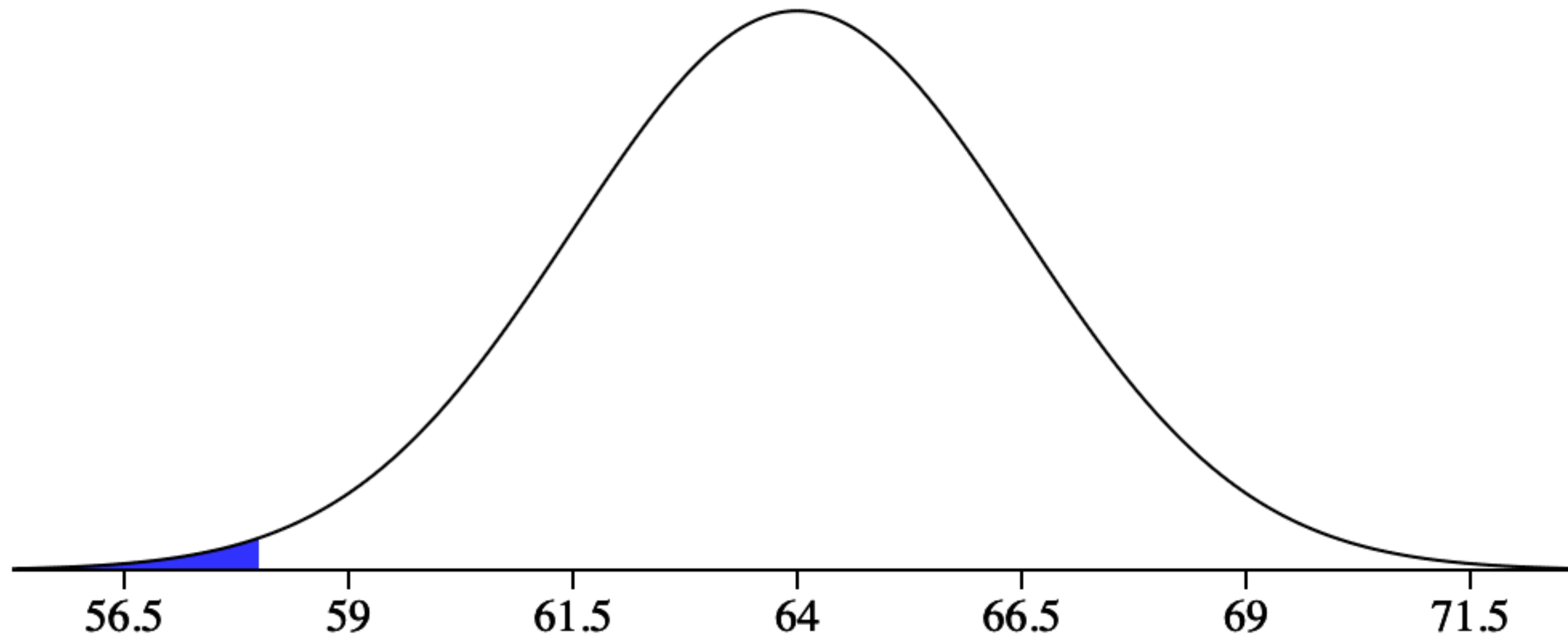
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is 64 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 58 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

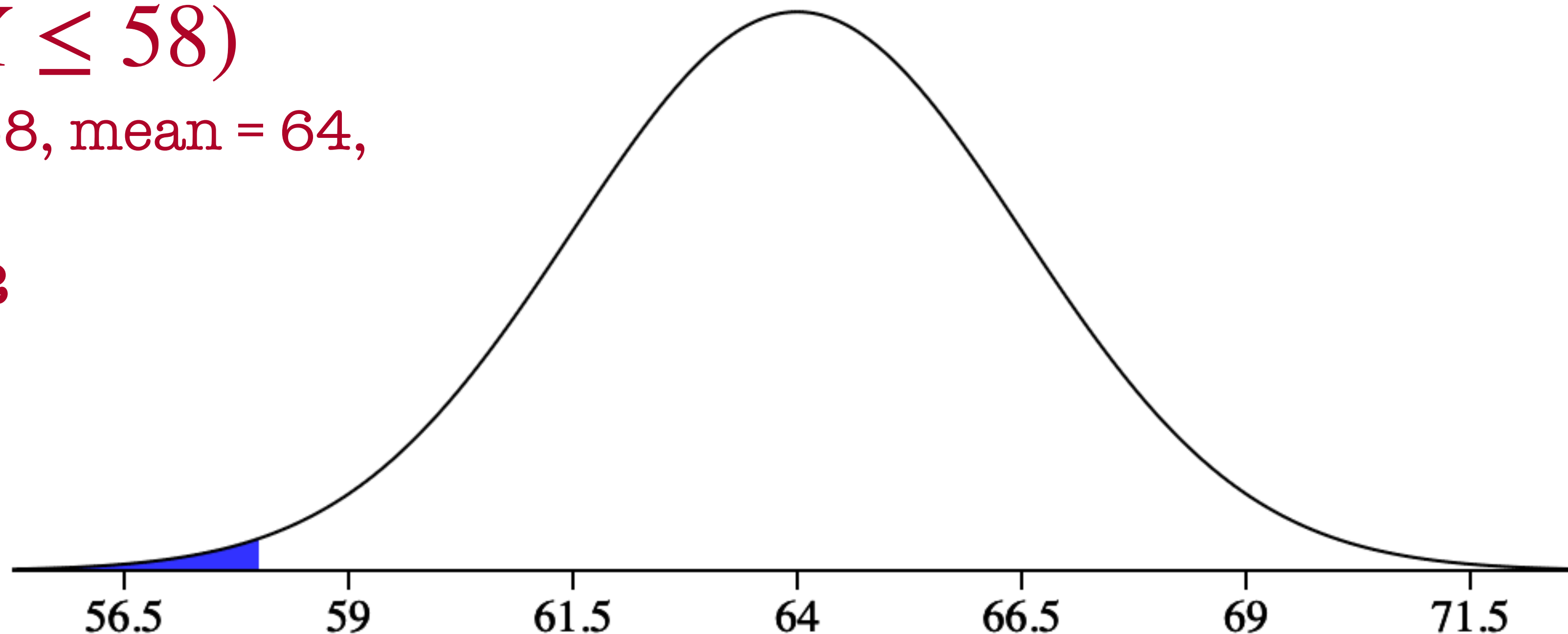
Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 58 inches tall?

$$P(X \leq 58)$$

`pnorm(58, mean = 64,
sd = 2.5)`

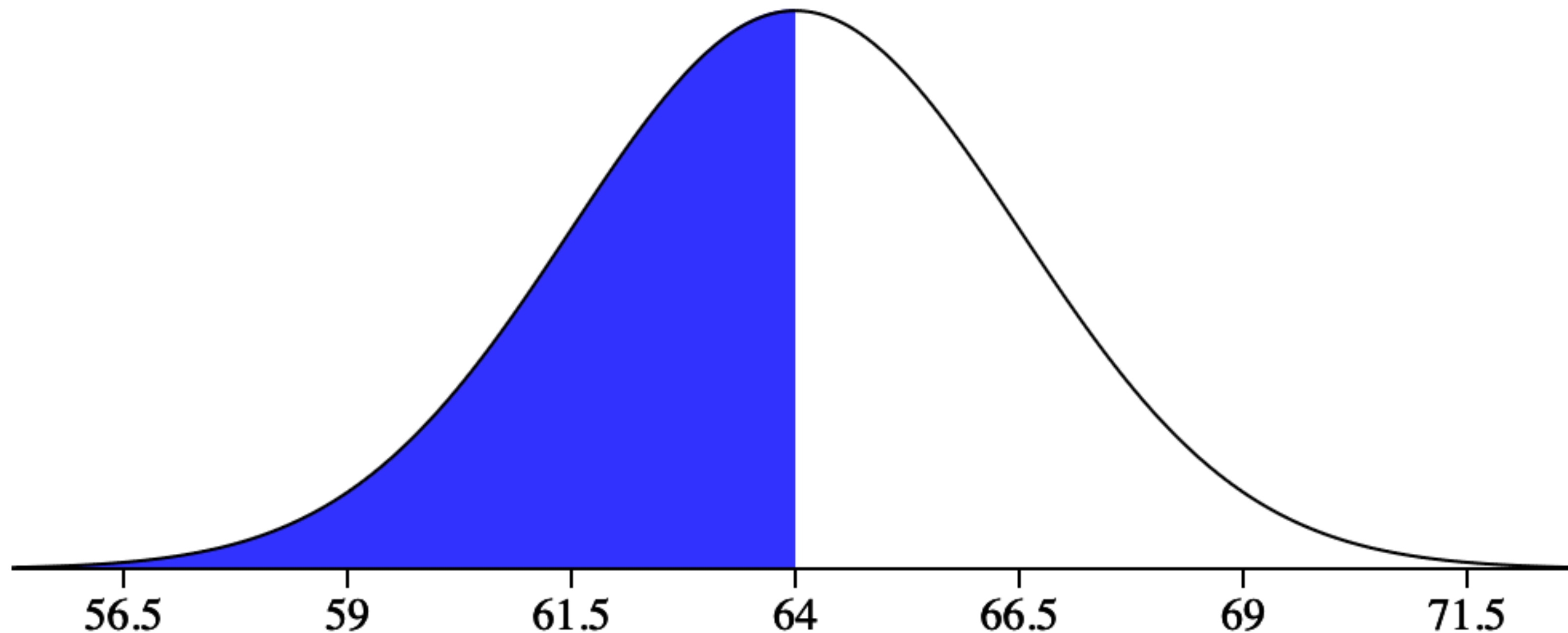
0.0082



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 64 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 64 inches tall?

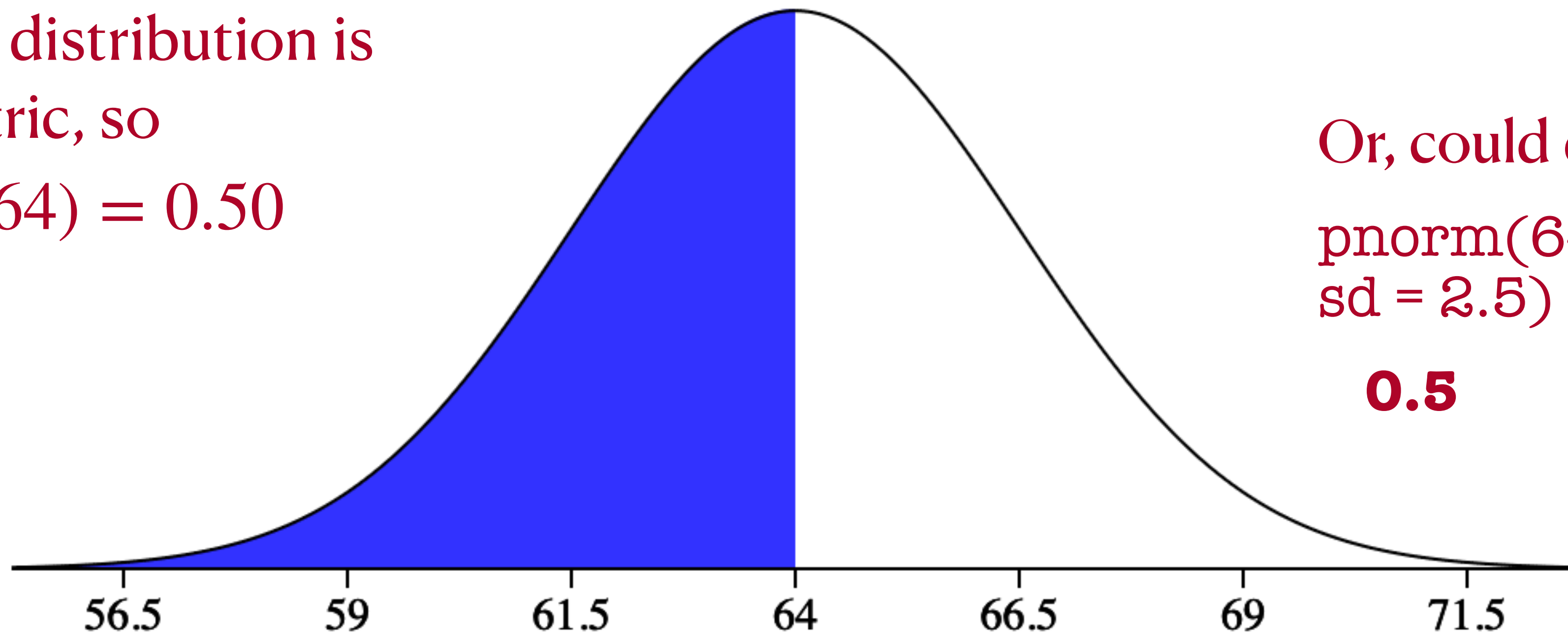
Normal distribution is
symmetric, so

$$P(X \leq 64) = 0.50$$

Or, could do...

`pnorm(64, mean = 64,
sd = 2.5)`

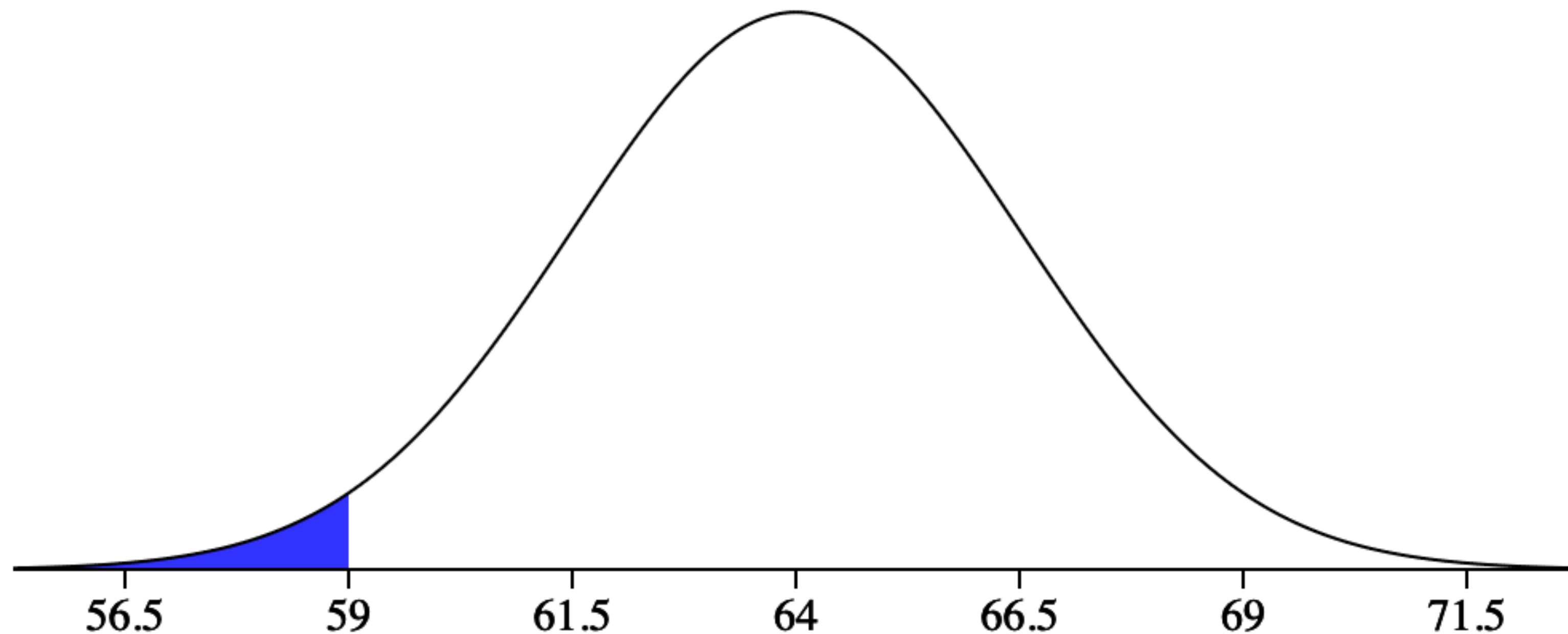
0.5



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 59 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is less than 59 inches tall?

If you don't need precise answer, could use empirical rule!

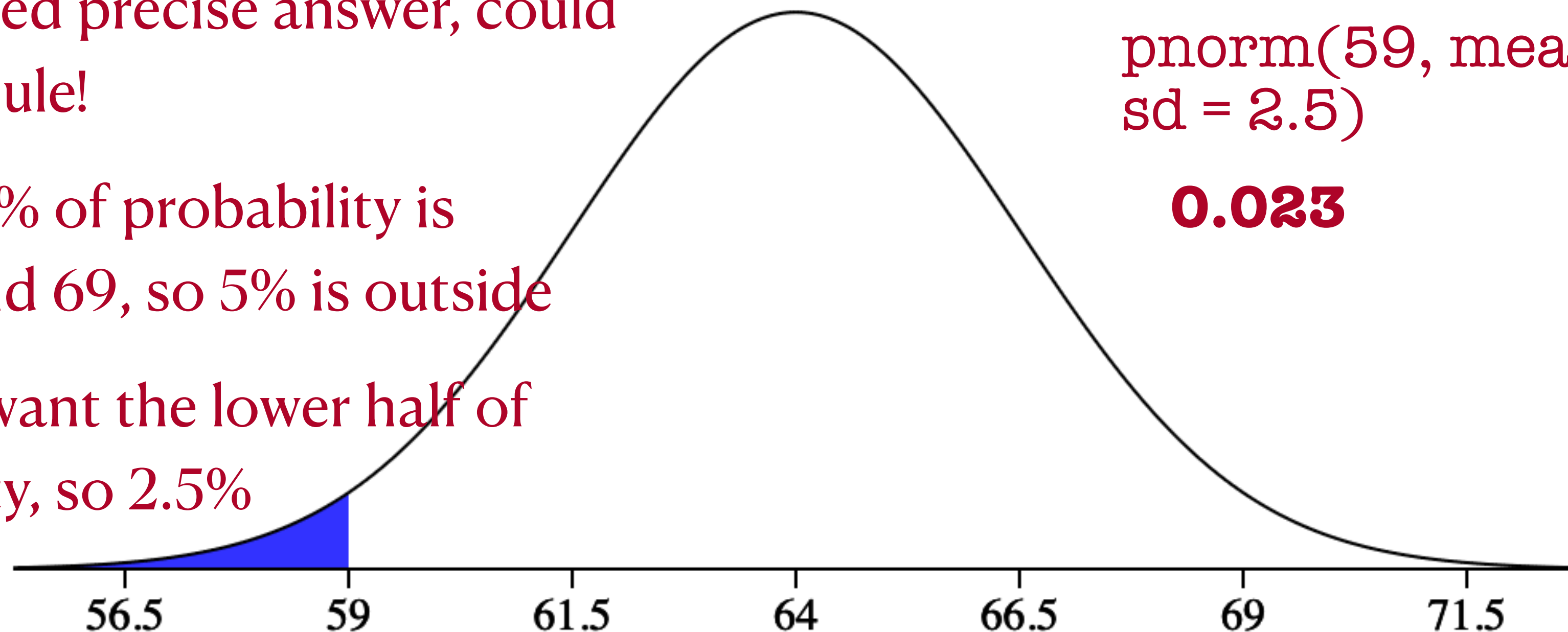
Said earlier 95% of probability is between 59 and 69, so 5% is outside

Here we only want the lower half of that probability, so 2.5%

Or, could do...

`pnorm(59, mean = 64, sd = 2.5)`

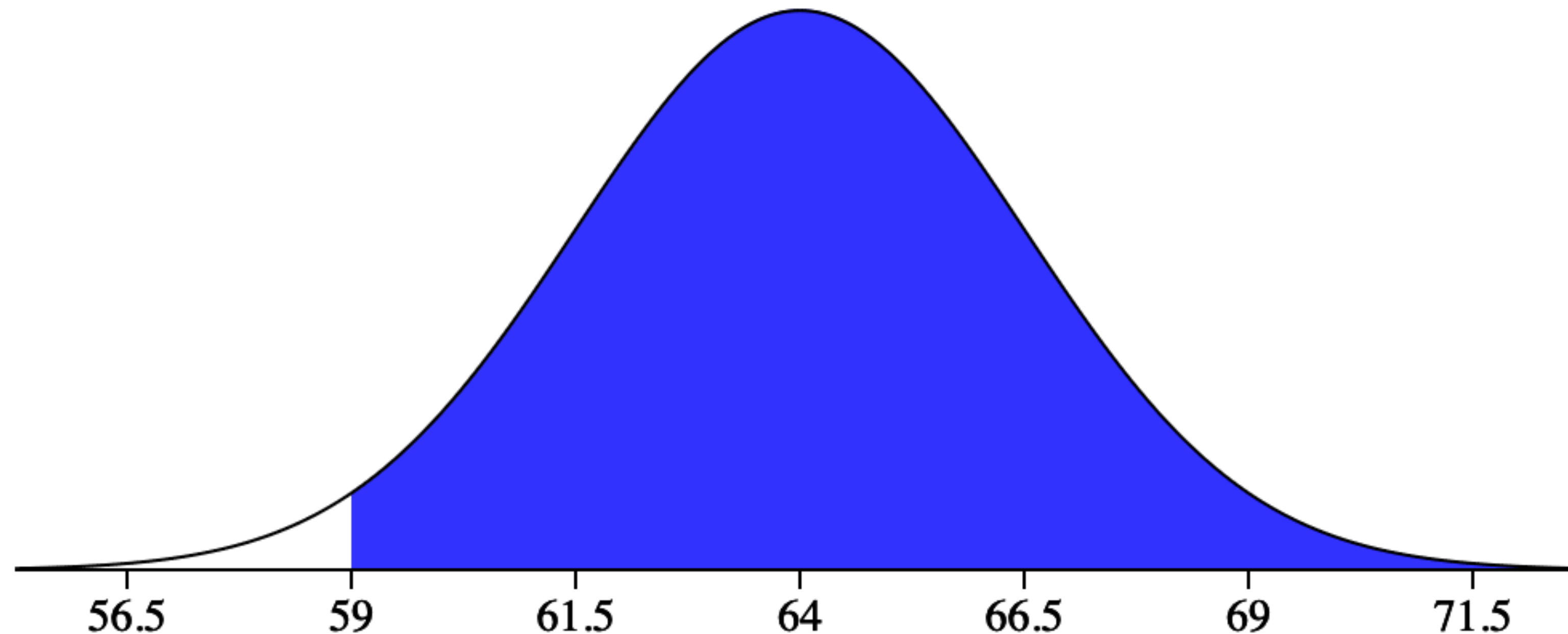
0.023



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is greater than 59 inches tall?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. What is the probability that a randomly selected US adult woman is greater than 59 inches tall?

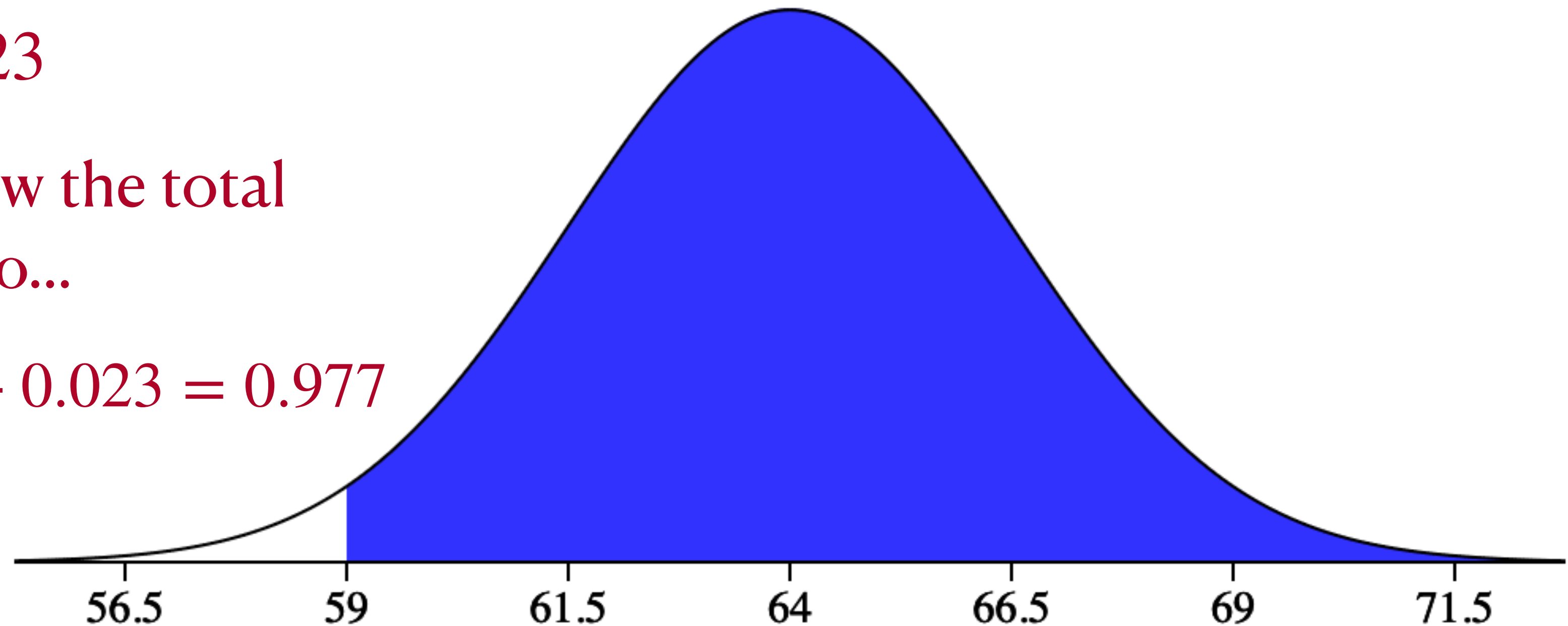
We just found $P(X \leq 59) = 0.023$

We want $P(X > 59)$, and we know the total area under the curve must be 1, so...

$$P(X > 59) = 1 - P(X \leq 59) = 1 - 0.023 = 0.977$$

Or, could do...

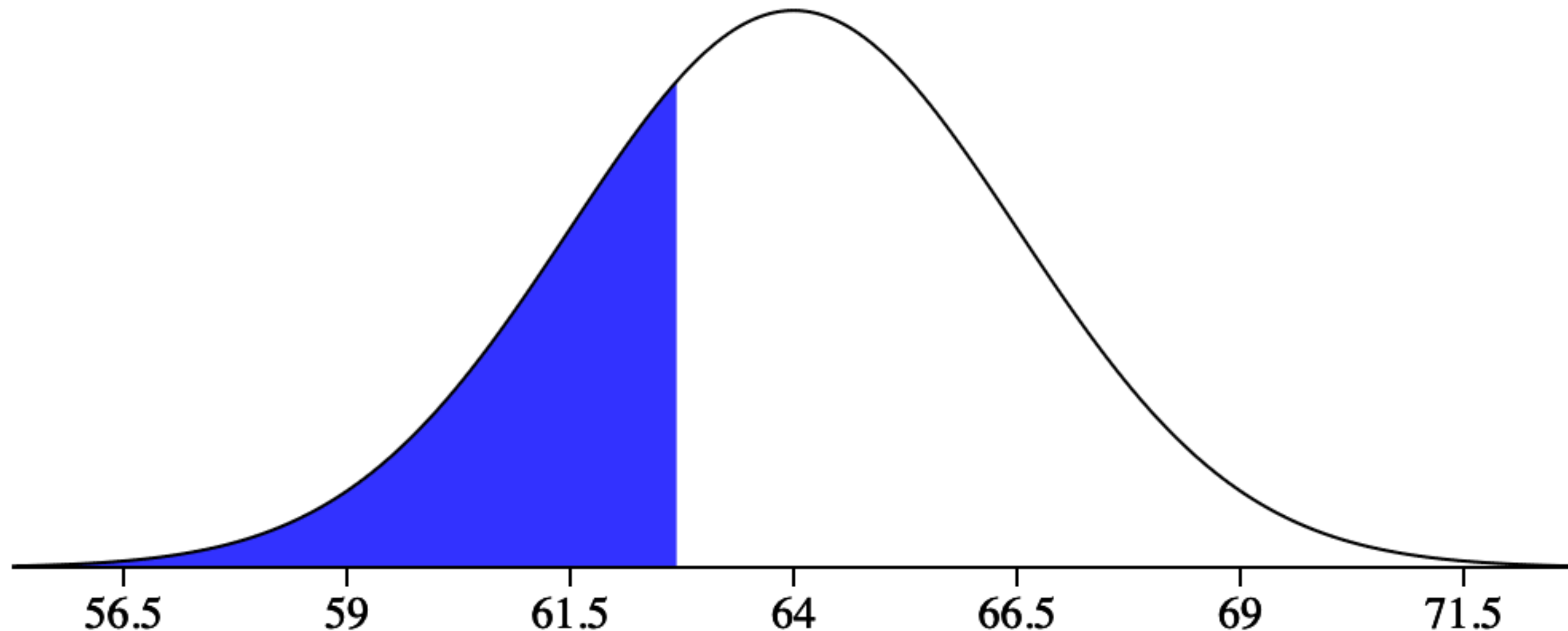
$$1 - \text{pnorm}(59, \text{mean} = 64, \text{sd} = 2.5) = \mathbf{0.977}$$



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is shorter than h is 0.30.



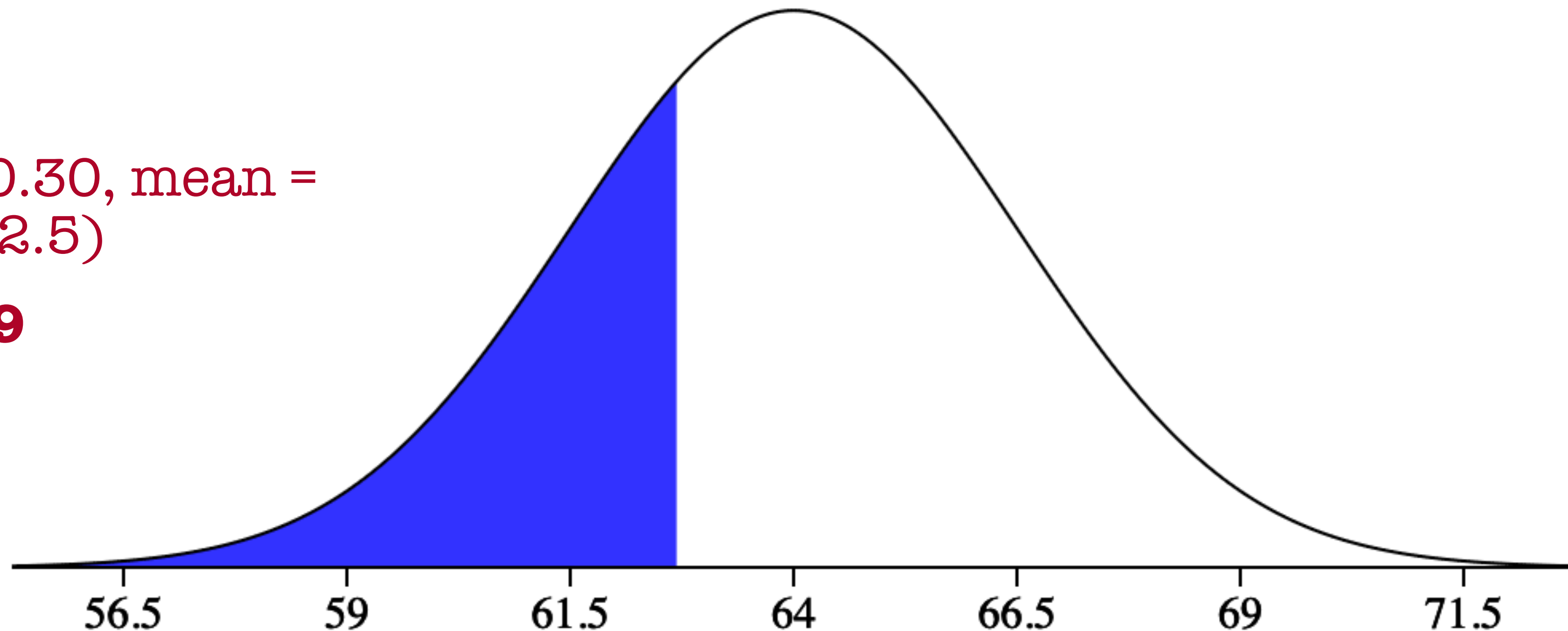
$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is shorter than h is 0.30.

`qnorm(0.30, mean =
64, sd = 2.5)`

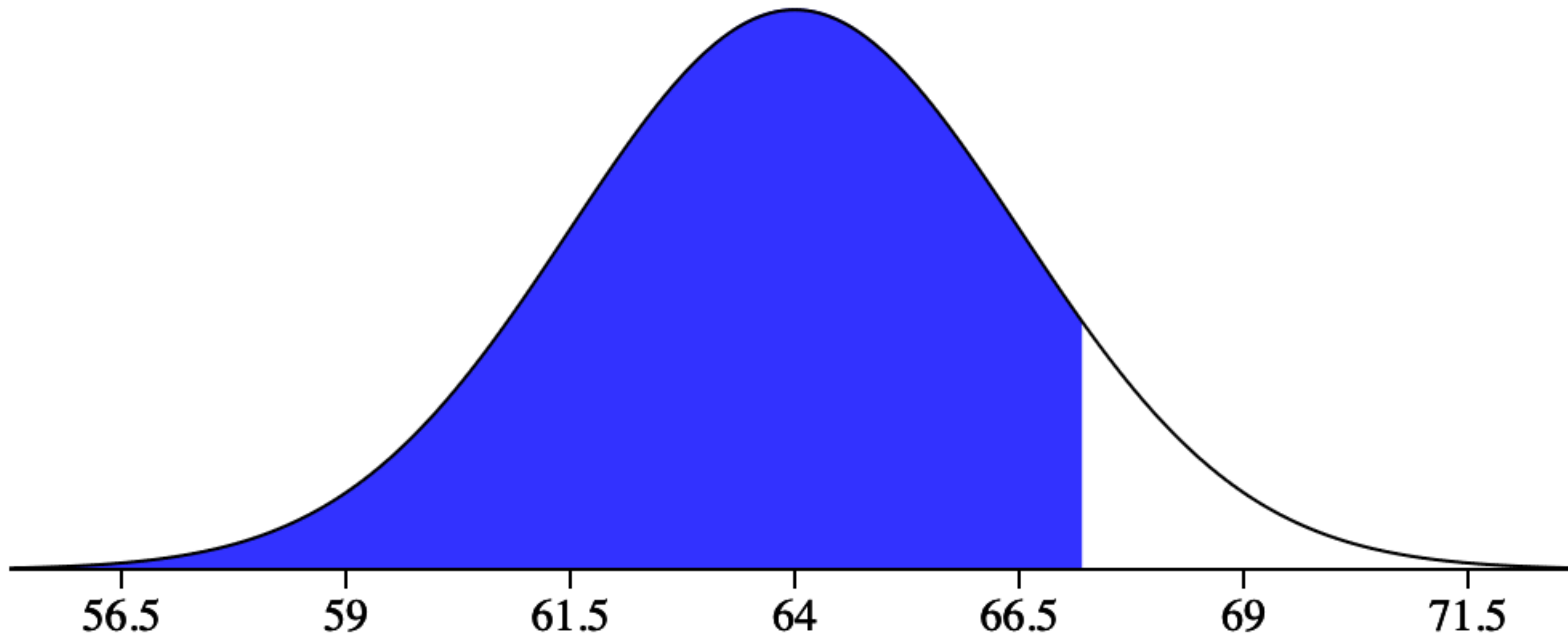
62.689



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is shorter than h is 0.90.



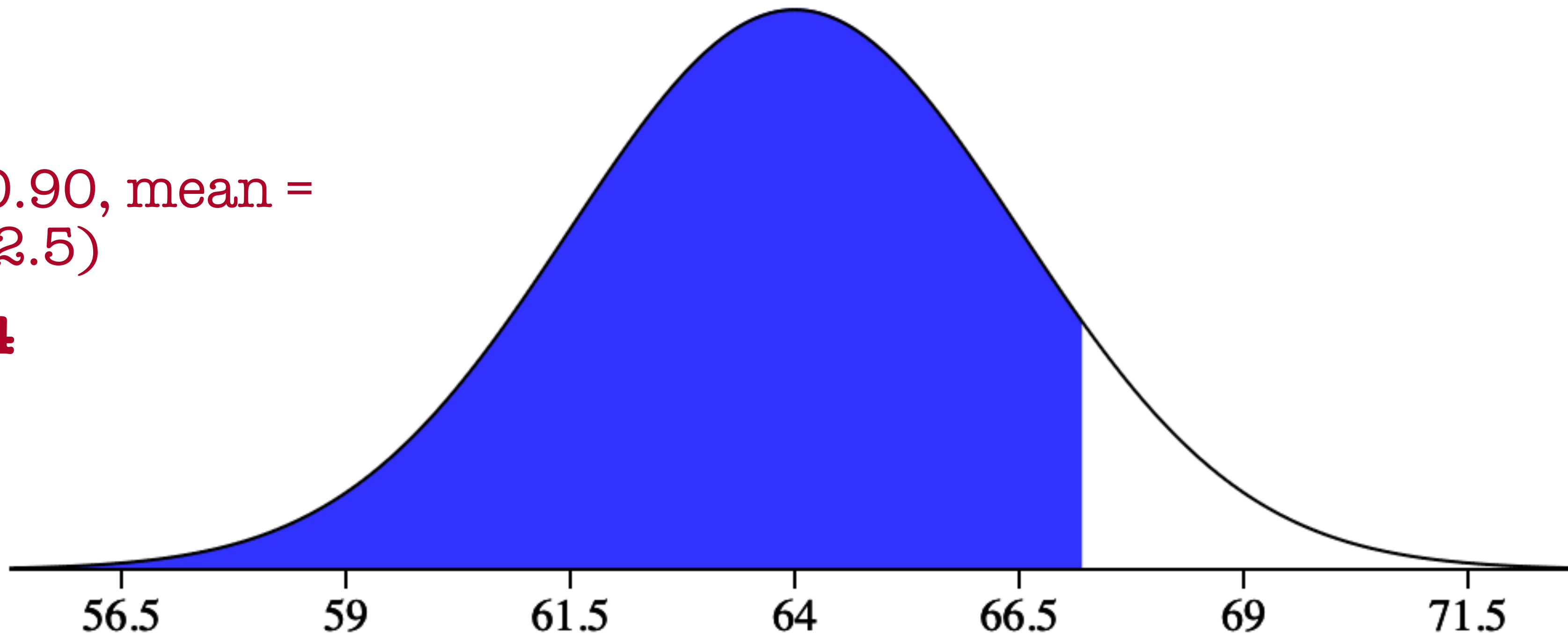
$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is shorter than h is 0.90.

`qnorm(0.90, mean =
64, sd = 2.5)`

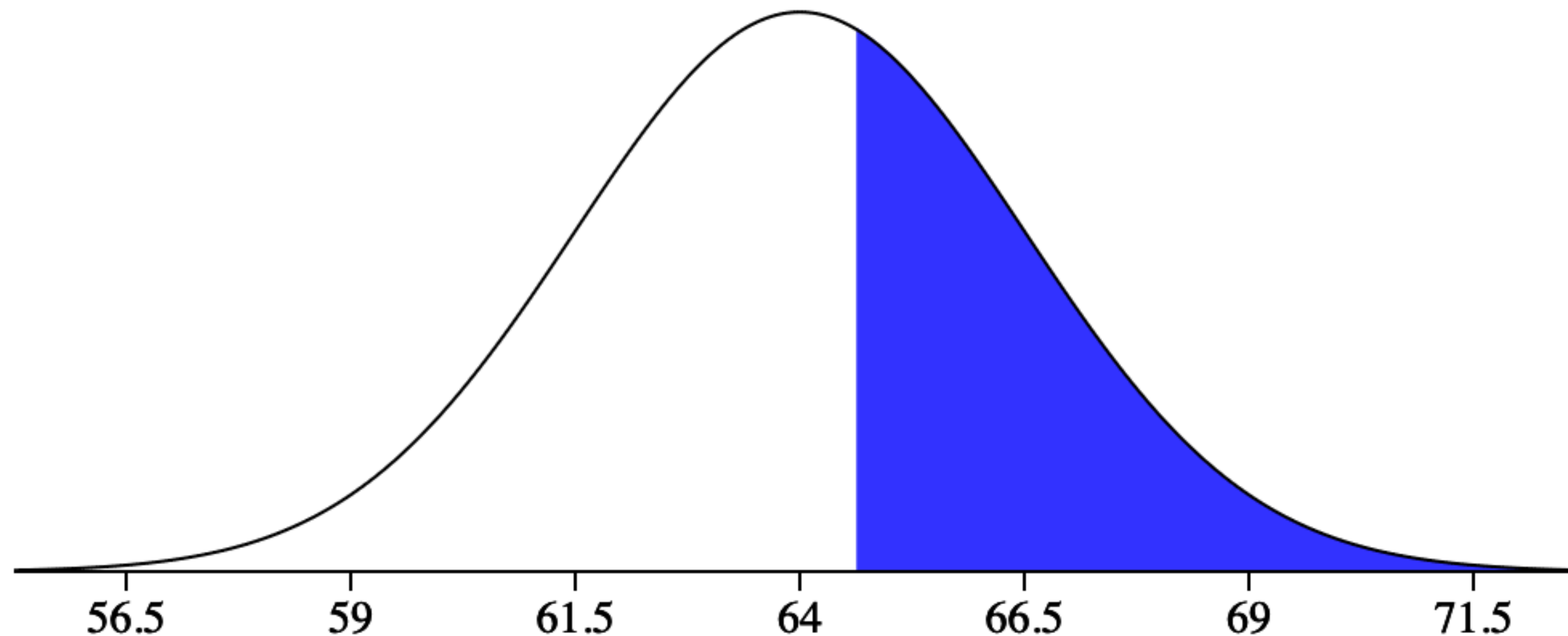
67.204



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is taller than h is 0.40.



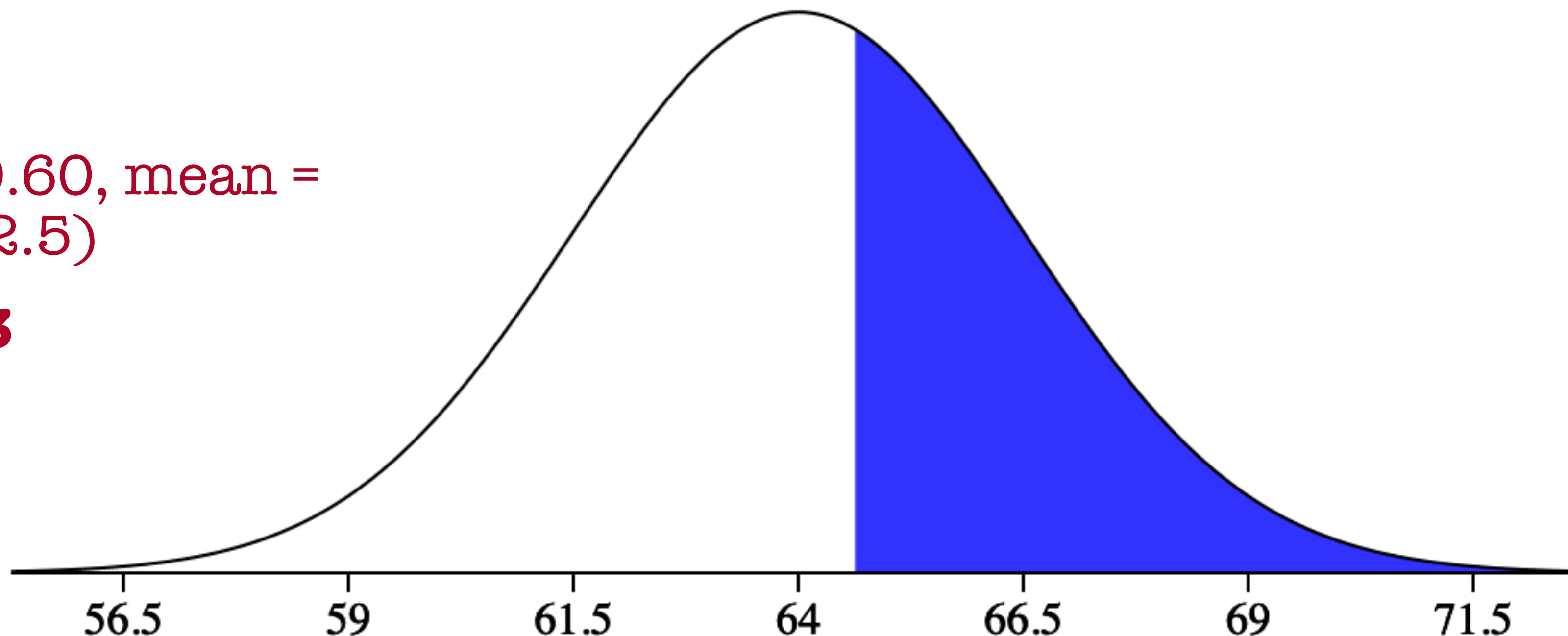
$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Normal distribution examples

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. Find the height h such that the probability a woman is **taller** than h is 0.40.

qnorm(0.60, mean =
64, sd = 2.5)

64.633



Z-Scores

(We'll cover this again in the next topic)

- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?

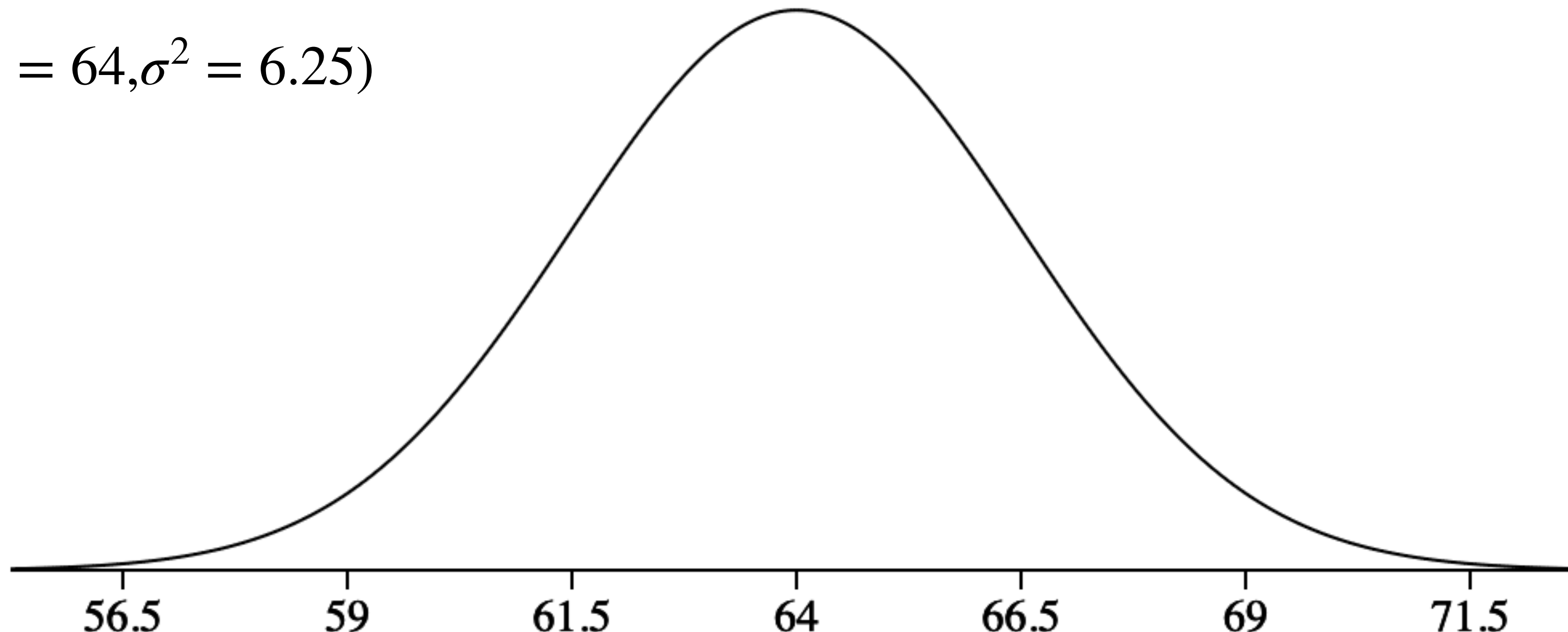
Z-Scores

(We'll cover this again in the next topic)

- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?

$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$

$\rightarrow \sigma = 2.5$



Z-Scores

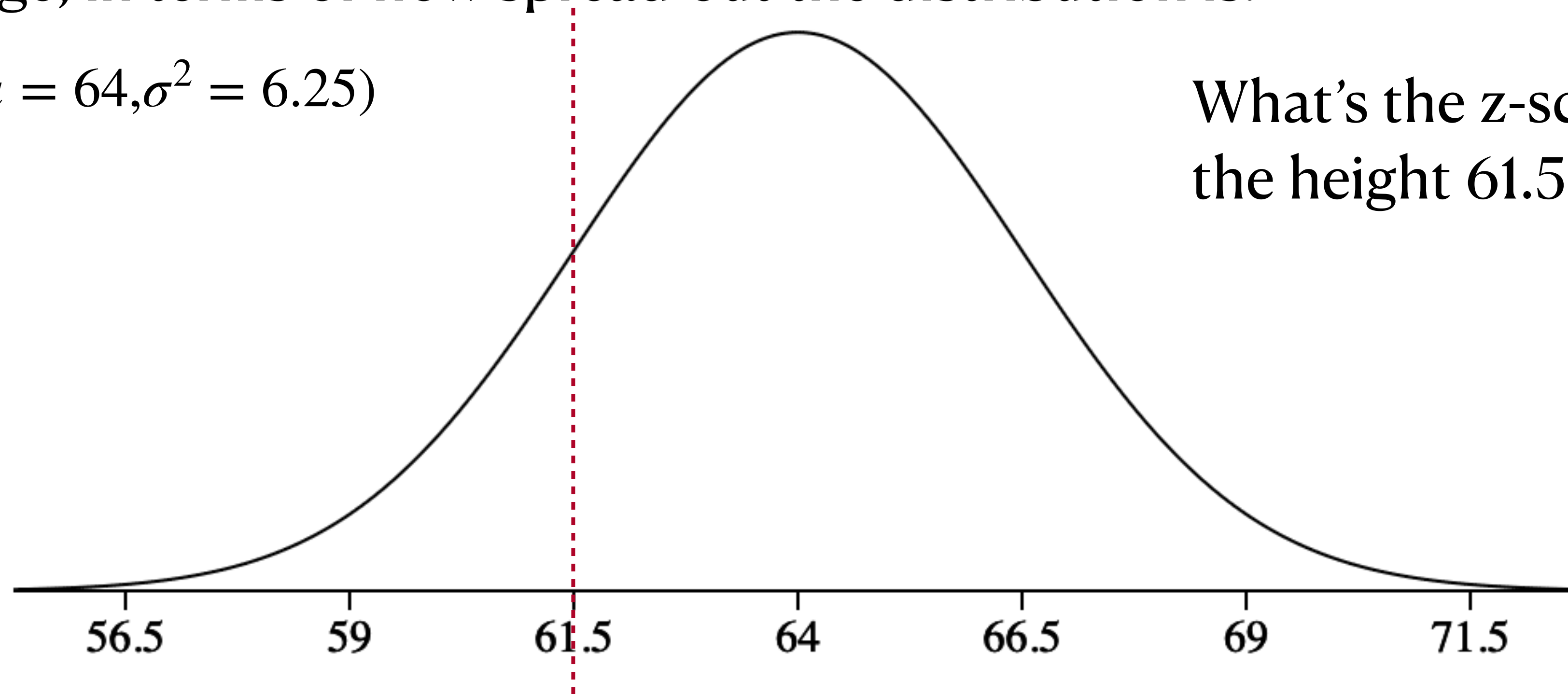
(We'll cover this again in the next topic)

- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?

$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$

$\rightarrow \sigma = 2.5$

What's the z-score for the height 61.5 inches?



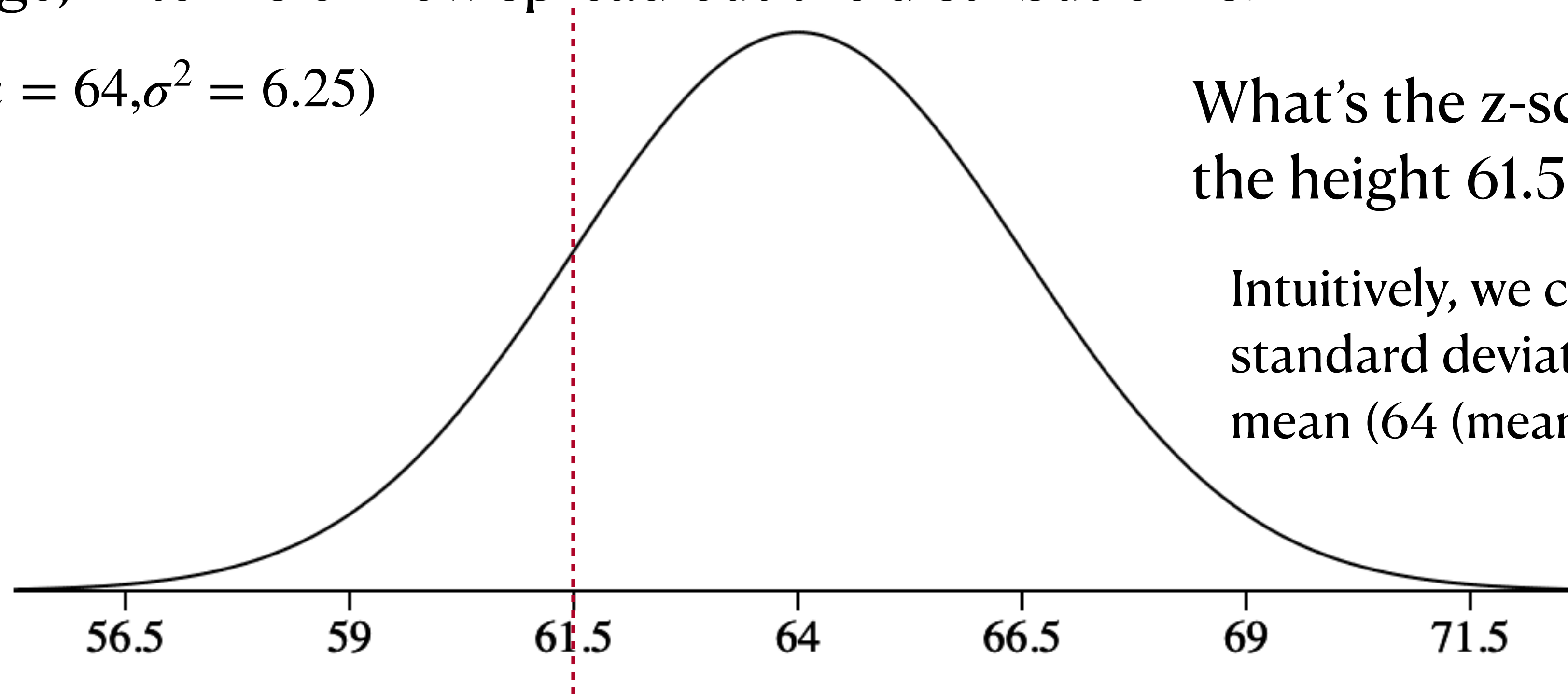
Z-Scores

(We'll cover this again in the next topic)

- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?

$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$

$\rightarrow \sigma = 2.5$



What's the z-score for the height 61.5 inches?

Intuitively, we can see that it's 1 standard deviation below the mean ($64 \text{ (mean)} - 2.5 \text{ (sd)} = 61.5$)

Z-Scores

(We'll cover this again in the next topic)

- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?

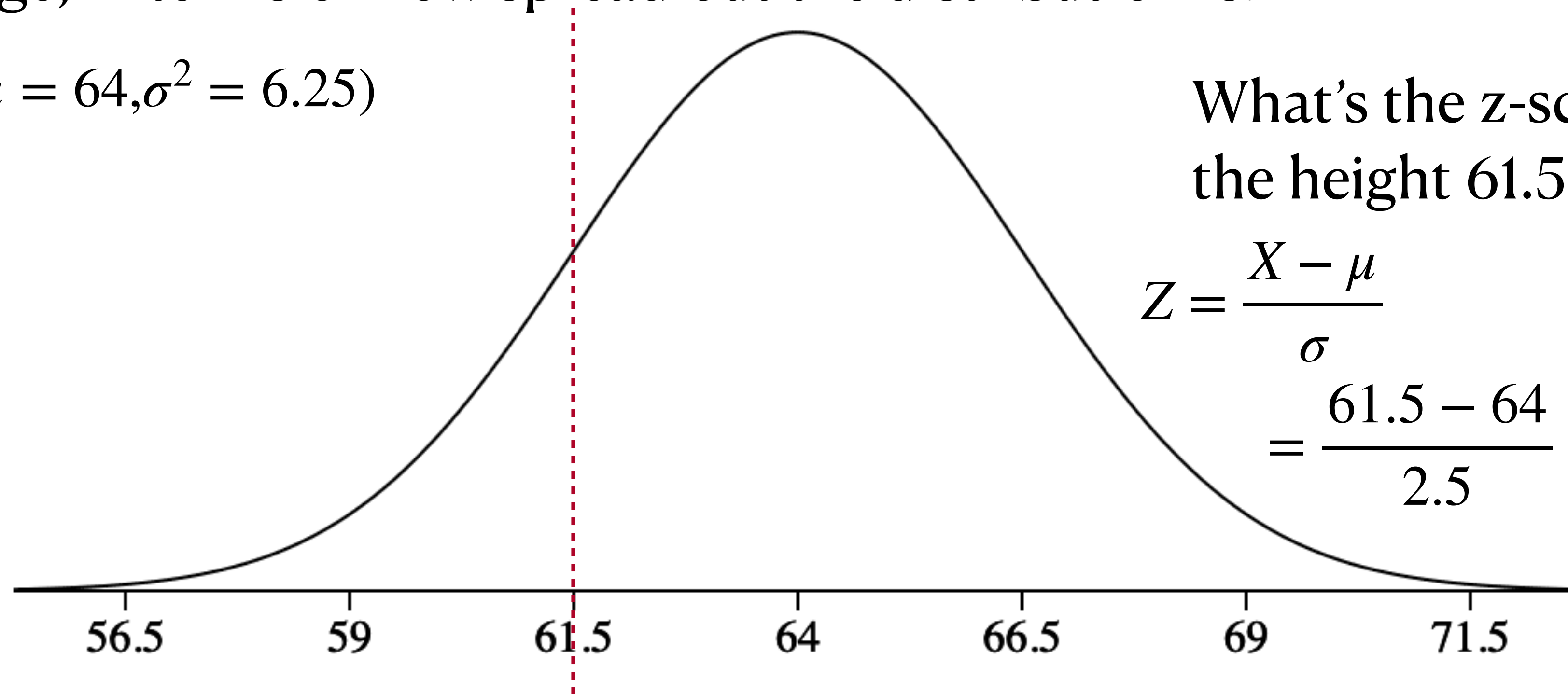
$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$

$\rightarrow \sigma = 2.5$

What's the z-score for the height 61.5 inches?

$$Z = \frac{X - \mu}{\sigma}$$

$$= \frac{61.5 - 64}{2.5} = \frac{-2.5}{2.5} = -1$$



Z-Scores

(We'll cover this again in the next topic)

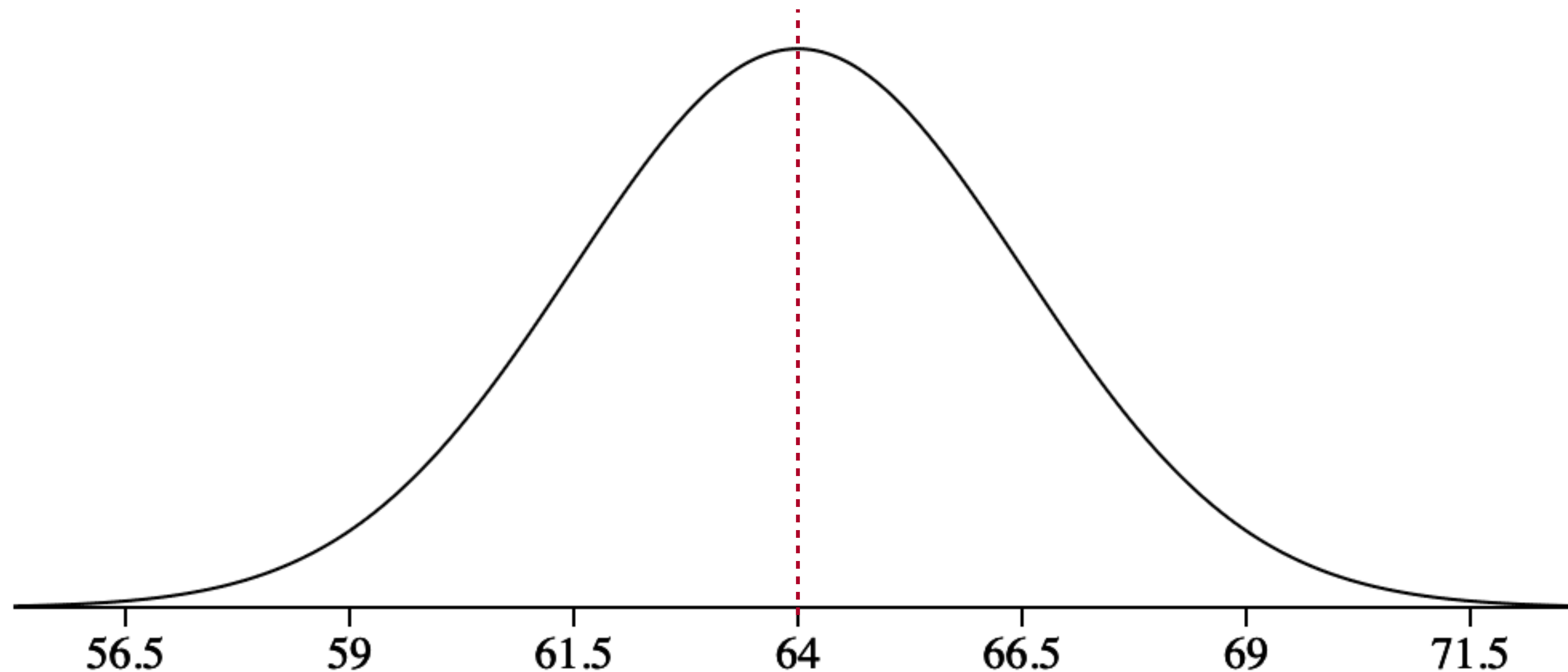
- The standardized score or **z-score** is a useful measure of the relative value of any observation in a dataset
- Counts the “number of standard deviations from the mean” — how far are you from the average, in terms of how spread out the distribution is?
- Computed by:

$$Z = \frac{(\text{value}) - (\text{mean})}{(\text{standard deviation})} = \frac{X - \mu}{\sigma}$$

$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Z-scores

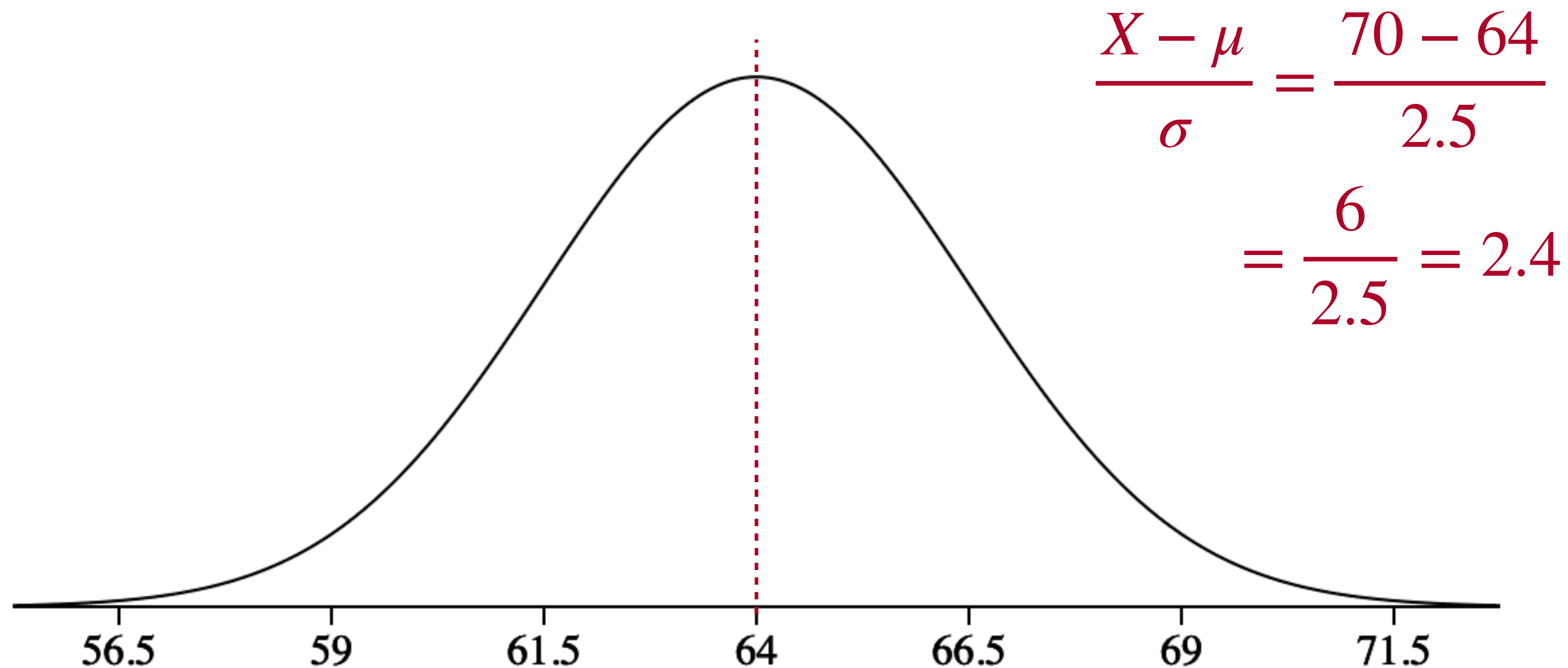
- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. A randomly selected woman is 70 inches tall. What is her z-score?



$$X \sim \text{Normal}(\mu = 64, \sigma^2 = 6.25)$$

Z-scores

- The distribution of heights (in inches) of adult women in the US can be modeled with this distribution. A randomly selected woman is 70 inches tall. What is her z-score?



Z-Scores: Why?

- A couple illustrative examples:
- Creatinine levels in blood (proxy for kidney function)
 - The normal creatinine level for a healthy person is 1.0 mg/dL. What if I go in and get my blood tested and I'm at 1.3 mg/dL? Seems not so bad until...
 - Standard deviation is 0.1 mg/dL. My z-score is 3, putting me at a very unusual/concerning value way above the mean!
- White blood cell count (measure of immune system activity)
 - Average healthy WBC is about 10,000 cells/ μ L. I go get tested, and I'm at 13,000. That's alarming, I'm really far above average! But...
 - Standard deviation is about 2000 cells/ μ L. My z-score is 1.5, putting me slightly high but nothing crazy.

★ **The raw difference only becomes meaningful once you scale it by how much variation is typical**